

Coordinate Geometry

ANSWER KEY



Distance and midpoint

- Find the distance between $A(1, 3)$ and $B(7, 11)$.
 $AB = \sqrt{(7-1)^2 + (11-3)^2} = \sqrt{36 + 64} = \sqrt{100} = 10.$
 - Find the midpoint of the segment joining $(-3, 5)$ and $(9, -1)$.
 $\text{Midpoint} = \left(\frac{-3+9}{2}, \frac{5+(-1)}{2}\right) = (3, 2).$
 - The midpoint of $A(2, -3)$ and $B(x, y)$ is $(5, 1)$. Find the coordinates of B .
 $\frac{2+x}{2} = 5$ gives $x = 8$. $\frac{-3+y}{2} = 1$ gives $y = 5$. So $B = (8, 5)$.
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Gradient and equations of lines

- Find the gradient of the line through $(2, -1)$ and $(6, 7)$.
 $\text{Gradient} = \frac{7 - (-1)}{6 - 2} = \frac{8}{4} = 2.$
 - Find the equation of the line through $(4, 3)$ with gradient -2 , in the form $y = mx + c$.
 $y - 3 = -2(x - 4)$, so $y = -2x + 8 + 3 = -2x + 11.$
 - Find the equation of the line through $(-1, 4)$ and $(3, -4)$, in the form $y = mx + c$.
 $\text{Gradient} = \frac{-4 - 4}{3 - (-1)} = \frac{-8}{4} = -2$. $y - 4 = -2(x + 1)$, so $y = -2x + 2.$
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Parallel and perpendicular lines

- Find the equation of the line parallel to $y = 3x - 1$ passing through $(5, 0)$.
Parallel lines share the same gradient, $m = 3$: $y - 0 = 3(x - 5)$, so $y = 3x - 15.$
 - Find the equation of the line perpendicular to $y = \frac{1}{3}x + 2$ passing through $(6, -1)$.
The perpendicular gradient is the negative reciprocal: $m = -3$. $y - (-1) = -3(x - 6)$, so $y = -3x + 18 - 1 = -3x + 17.$
 - Determine whether the lines $y = 4x - 3$ and $4y + x = 8$ are parallel, perpendicular, or neither.
Rewrite the second line as $y = -\frac{1}{4}x + 2$, gradient $-\frac{1}{4}$. The product of the gradients is $4 \times \left(-\frac{1}{4}\right) = -1$, so the lines are perpendicular.
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The equation of a circle

- 10** Write the equation of a circle with center $(3, -2)$ and radius 6.
 $(x - 3)^2 + (y + 2)^2 = 36$.
- 11** Find the center and radius of $x^2 + y^2 - 4x + 6y - 12 = 0$ by completing the square.
 Group: $(x^2 - 4x) + (y^2 + 6y) = 12$. Complete the square: $(x - 2)^2 - 4 + (y + 3)^2 - 9 = 12$, so
 $(x - 2)^2 + (y + 3)^2 = 25$. Center: $(2, -3)$. Radius: $\sqrt{25} = 5$.
- 12** A circle has equation $(x + 1)^2 + (y - 4)^2 = 49$. State its center and radius.
 Center: $(-1, 4)$. Radius: $\sqrt{49} = 7$.

Lines and circles together

- 13** A circle has center $(0, 0)$ and radius 5. Determine whether the line $y = x + 8$ intersects the circle, is tangent to it, or misses it entirely, by comparing the distance from the center to the line against the radius.
 Rewrite the line as $x - y + 8 = 0$. Distance from $(0, 0)$: $d = \frac{|0 - 0 + 8|}{\sqrt{1^2 + (-1)^2}} = \frac{8}{\sqrt{2}} = 4\sqrt{2} \approx 5.66$. Since $5.66 > 5$ (the distance exceeds the radius), the line misses the circle entirely.
- 14** A circle has center $(2, 1)$ and radius 3. Determine whether the line $x + y = 10$ intersects the circle, is tangent to it, or misses it entirely.
 Rewrite the line as $x + y - 10 = 0$. Distance from $(2, 1)$: $d = \frac{|2 + 1 - 10|}{\sqrt{1^2 + 1^2}} = \frac{7}{\sqrt{2}} = \frac{7\sqrt{2}}{2} \approx 4.95$. Since $4.95 > 3$, the line misses the circle entirely.

The perpendicular bisector

- 15** Find the equation of the perpendicular bisector of the segment joining $A(1, 2)$ and $B(7, 8)$.
 Midpoint: $(4, 5)$. Gradient of AB : $\frac{8 - 2}{7 - 1} = 1$. Perpendicular gradient: -1 . Equation: $y - 5 = -1(x - 4)$, so $y = -x + 9$.
- 16** Find the equation of the perpendicular bisector of the segment joining $A(-2, 3)$ and $B(4, -5)$.
 Midpoint: $(1, -1)$. Gradient of AB : $\frac{-5 - 3}{4 - (-2)} = \frac{-8}{6} = -\frac{4}{3}$. Perpendicular gradient: $\frac{3}{4}$. Equation:
 $y + 1 = \frac{3}{4}(x - 1)$, so $y = \frac{3}{4}x - \frac{7}{4}$.

Area of a triangle from coordinates

- 17** Find the area of the triangle with vertices $P(0, 0)$, $Q(8, 0)$, $R(3, 6)$.
 Using the coordinate area formula: Area
 $= \frac{1}{2}|x_P(y_Q - y_R) + x_Q(y_R - y_P) + x_R(y_P - y_Q)| = \frac{1}{2}|0(0 - 6) + 8(6 - 0) + 3(0 - 0)| = \frac{1}{2}(48) = 24$.

- 18 Find the area of the triangle with vertices $A(-3, 2)$, $B(5, 2)$, $C(1, 7)$.

Area

$$= \frac{1}{2} |x_A(y_B - y_C) + x_B(y_C - y_A) + x_C(y_A - y_B)| = \frac{1}{2} |(-3)(2 - 7) + 5(7 - 2) + 1(2 - 2)| = \frac{1}{2} |15 + 25 + 0| = \frac{1}{2} (40) = 20$$

A well-designed word problem: locating a cell tower

- 19 This question has two parts. Two houses are located at $A(0, 0)$ and $B(12, 0)$ (positions in kilometers). A cell tower must be built equidistant from both houses, and it must also lie on the line $y = x + 2$. (a) Find the equation of the perpendicular bisector of AB , and use it together with $y = x + 2$ to find the tower's coordinates. (b) Find the distance from the tower to each house, and confirm they are equal.

(a) Midpoint of AB : $(6, 0)$. Since AB is horizontal, its perpendicular bisector is the vertical line $x = 6$.

Substituting into $y = x + 2$: $y = 8$. Tower location: $(6, 8)$. (b) Distance to A :

$$\sqrt{(6 - 0)^2 + (8 - 0)^2} = \sqrt{36 + 64} = \sqrt{100} = 10 \text{ km. Distance to } B:$$

$$\sqrt{(6 - 12)^2 + (8 - 0)^2} = \sqrt{36 + 64} = \sqrt{100} = 10 \text{ km. Both distances equal 10 km, confirming the tower is equidistant from both houses.}$$